



Scientific Committee on Consumer Safety

SCCS

SCIENTIFIC ADVICE on

on Silver used in cosmetic products

(CAS/EC No. 7440-22-4/231-131-3)



The SCCS adopted this document written procedure on 17 December 2025

ACKNOWLEDGMENTS

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1. ABSTRACT

The SCCS concludes the following:

1. *In light of the new data provided, the previous SCCS Opinion (SCCS/1665/24) and taking under consideration the classification as toxic for reproduction Cat. 2, does the SCCS consider micron-sized particulate Silver [100 nm < particle diameter < 1 mm] safe when used up to a maximum concentration of 0.2 % in rinse-off and 0.3 % in leave-on cosmetic products?*

In the light of the new data provided, indicating that micron-sized particulate Silver does not penetrate the skin, the SCCS considers micron-sized particulate Silver [100nm<particle diameter<1 mm] safe when used up to a maximum concentration of 0.2% in rinse-off and 0.3% in leave-on cosmetic products, as specified in the Tables 5 and 6.

This Opinion does not apply to the use of micron-sized particulate Silver [100nm<particle diameter<1 mm] in propellant-based spray products because such applications were not covered by the Applicant, and therefore not assessed for safety by the SCCS.

2. *Alternatively, what is according to the SCCS, the maximum concentration considered safe for use of micron-sized particulate Silver [100 nm < particle diameter < 1 mm] in cosmetic products?*

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3. *Does the SCCS have any further scientific concerns with regard to the use of micron-sized particulate Silver [100 nm < particle diameter < 1 mm] in cosmetic products?*

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Keywords: SCCS, scientific advice, Silver, Regulation 1223/2009, CAS/EC No. 7440-22-4/231-131-3

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SCCS

The Committee shall provide Opinions on questions concerning health and safety risks (notably chemical, biological, mechanical and other physical risks) of non-food consumer products (for example cosmetic products and their ingredients, toys, textiles, clothing, personal care and household products such as detergents, etc.) and services (for example: tattooing, artificial sun tanning, etc.).

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2. MANDATE FROM THE EUROPEAN COMMISSION

Background

Silver (CAS/EC No. 7440-22-4/231-131-3) is an ingredient primarily used as a colorant (CI 77820) in cosmetics, providing a silver hue to various cosmetic formulations. It is an authorised colorant and, therefore, listed in entry 142 of Annex IV to the Cosmetics Regulation. Silver is frequently found in makeup products such as eyeshadows, highlighters, nail polishes, and body powders, where it provides a metallic/shimmery effect. In addition, in the current dossier submission, silver is reported as conditioning agent in rinse-off and leave on cosmetic products.

The European Risk Assessment Committee (RAC) of ECHA issued in February 2023 an opinion recommending among others a 'Toxic for Reproduction Category 2' classification for Silver. Following the RAC opinion, the European Commission may propose a classification for silver as a 'Toxic for Reproduction Category 2' (CLP Regulation Annex VI entry).

According to Article 15(1) of the Cosmetics Regulation 'the use in cosmetic products of substances classified as CMR substances, of category 2, under Part 3 of Annex VI to Regulation (EC) No 1272/2008 shall be prohibited. However, a substance classified in category 2 may be used in cosmetic products where the substance has been evaluated by the SCCS and found safe for use in cosmetic products'. In view of these provisions, regulatory measures must be adopted by the Commission services within 15 months of the classification as CMR 1A or 1B of the substance(s) concerned in Part 3 of Annex VI to Regulation (EC) No 1272/2008.

In October 2023, the Commission services received a dossier to defend the safe use of micron-sized particulate Silver (CAS/EC No. 7440-22-4/231-131-3) as a conditioning agent in cosmetic products according to Article 15(1) of the Cosmetics Regulation 1223/2009.

In June 2024, the SCCS concluded in its Opinion (SCCS/1665/24) that micron-sized particulate Silver was not safe at concentrations up to 0.2 % in rinse-off and 0.3 % in leave-on cosmetic products when used alone or in combination. Nevertheless, the SCCS noted that the use of micron-sized particulate Silver in eye shadow and oral exposure products (lip balm, toothpaste and mouthwash) at concentrations mentioned in Section 3.5 of the Opinion was considered safe, either used alone or in combination.

In August 2025, the Commission services received an updated dossier focusing on dermal penetration studies (ex vivo and human volunteer study), aiming to demonstrate the safe use of micron-sized particulate Silver. The Commission, therefore, requests the SCCS to carry out a safety assessment on this ingredient in view of the new information provided.

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Terms of reference

1. *In light of the new data provided, the previous SCCS Opinion (SCCS/1665/24) and taking under consideration the classification as toxic for reproduction Cat. 2, does the SCCS consider micron-sized particulate Silver [100 nm < particle diameter < 1 mm] safe when used up to a maximum concentration of 0.2 % in rinse-off and 0.3 % in leave-on cosmetic products?*
2. *Alternatively, what is according to the SCCS, the maximum concentration considered safe for use of micron-sized particulate Silver [100 nm < particle diameter < 1 mm] in cosmetic products?*
3. *Does the SCCS have any further scientific concerns with regard to the use of micron-sized particulate Silver [100 nm < particle diameter < 1 mm] in cosmetic products?*

3. SCIENTIFIC ADVICE

Preamble

This Scientific Advice has a focus on the exposure assessment and subsequent safety evaluation of micron-sized particulate Silver used in cosmetic products, in the light of the newly submitted data. The toxicological aspects of the material under evaluation have already been assessed in a recent SCCS Opinion (SCCS/1665/24, dated 20 June 2024). Whereas the newly submitted data concerned the skin penetration of micron-sized particulate Silver, in this Scientific Advice, the SCCS also considered the dermal absorption of silver ions that are released from the particles.

3.1 CHEMICAL AND PHYSICAL SPECIFICATIONS

3.1.1 Chemical identity

Silver

3.1.1.3 Trade names and abbreviations

From Applicant:
MicroSilver BG™ (referred to as 'MicroSilver BG')

3.1.1.4 CAS / EC number

From Applicant: CAS No. 7440-22-4/ EC No. 231-131-3

3.1.2 Physical form

(Partially taken from SCCS/1665/24 and abstracted from the Applicant's previous dossier)

The particles could be described as porous and sintered. The pre-treatment with sonification for 35 minutes shows that the particles cannot be broken down into smaller parts. According to the manufacturer, it was not possible to chop the particles down to a smaller size. High forces only led to a compression of the particles and a reduction of the surface area.

The number-based particle size of the unbound test substance with individual existence ranged between 0.13 and 20.69 µm.

The mean (±SD) measured particle size was 2.35 ± 3.38 µm.

The number-based particle size distribution was characterised by specific percentiles: D10 at 0.22 µm, D50 at 1.28 µm, and D90 at 5.46 µm.

Ref.: SCCS/1665/24

Table 1. Summary of the results from both SEM and TEM analysis for the test substance and reference substance.

Test substance	EM	Name	Chemical name	Particle size / d ₀ , ecd*	<100 nm	Counted particles
Test substance, dispersion, substructures (I)	SEM	MicroSilver BG	Silver	D10: 80.5 nm D50: 116.2 nm D90: 172.8 nm	377	1288
Test substance, dispersion, aggregates (II) (unbound basic MicroSilver BG units)	SEM	MicroSilver BG	Silver	D10: 220 nm D50: 1280 nm D90: 5460 nm	0	75
Test substance, dispersion, substructures (I)	TEM	MicroSilver BG	Silver	D10: 65.5 nm D50: 106 nm D90: 173.9 nm	222	504
Reference substance(I)	SEM	RM 8017	Polyvinylpyrrolidone coated Silver	D10: 68.1 nm D50: 75.7 nm D90: 81.3 nm	520	520
Reference substance	TEM	RM 8017	Polyvinylpyrrolidone coated Silver	D10: 58.1 nm D50: 66.7 nm D90: 73.6 nm	485	485

ecd* = equivalent circular diameter

Based on the generated data, it can be concluded that according to the EC nanodefinition recommendation 2011/696 as well as the new recommendation 2022/C 229/01, MicroSilver BG is not a nanomaterial because it does not fulfil the following conditions:

Individual existence or existing as identifiable particles in aggregates or agglomerates, where 50% or more of these particles in the numerical size distribution meet at least one of the following conditions.

- one or more external dimensions of the particle are in the size range of 1 nm to 100 nm;
- the particle has an elongated shape, such as a rod, fibre, or tube, in which two external dimensions are less than 1 nm and the other dimension is greater than 100 nm;
- the particle has a plate-like shape in which one outer dimension is less than 1 nm and the other is greater than 100 nm.

The basic units of MicroSilver BG particles, which have unbound individual existence, are not nanomaterials as their external dimensions clearly exceed 100 nm.

In accordance with the definitions recommended by EC 2022/C 229/01 (see below), it is appropriate to consider MicroSilver BG unbound particles as 'particles' rather than 'aggregates.' This distinction is made because, during the production process, there are no free particles or constituent particles that bind together to form aggregates.

SCCS comment

According to SCCS Guidance on nanomaterials in cosmetics (SCCS/1655/23), a batch-to-batch variation of at least 5 batches should have been provided.

Appropriate HR-SEM and HR-TEM images should have been provided indicating representative examples of how the dimension of the substructures (constituent particles) and the particles (aggregates) have been measured.

Based on the submitted documentation, the SCCS agrees that micron-sized particulate Silver is not a nanomaterial.

3.1.3 Homogeneity and Stability: ionisation

From SCCS/1665/24 and its underlying documents submitted by the Applicant:

According to the Applicant, the microparticles offer a highly biocompatible depot of pure silver that provides sustainable release of silver ions. The sponge-like particle structure forms a relatively high surface area – in comparison to usual silver powder – that allows the sustainable generation of silver ions at low concentrations over a longer period of time.

Table 2 (taken from Appendix A in SCCS/1665/24):

Applicant's Table with results showing the release of Silver-ions from MicroSilver BG in different formulations. Percentage of Silver ions in various cosmetic formulations containing MicroSilver BG measured by Inverse voltammetry which selectively detect Silver ions.

Product	MicroSilver BG content (Ag ₀) [%]	Silver ions (Ag ⁺) µg/g in the cosmetic product	CV% *	[%] of total Silver ion (Ag ⁺) in the cosmetic product
MicroSilver BG Skin ointment	0.1	0.308	43.8	0.03
MicroSilver BG soothing shampoo	0.1	0.467	11.7	0.05
SOS MicroSilver cream	0.1	0.885	8.7	0.09
Diabetic Feet Foam Formulation with MicroSilver BG (after 3 years of storage)	0.2	0.001	1.0	0.00005
Diabetic Feet Foam Formulation with MicroSilver BG (freshly manufactured)	0.2	0.003	2.3	0.15

*coefficient of variation

Applicant's Conclusion:

Taking into account the available silver ions in the cosmetic formulations of less than 0.1 %, the available silver ions of 0.001 % in the study by Mertens et al. from 2023 and the above release kinetics for silver ions under ideal conditions, a very conservative ionization rate of 0.5% is obtained. This 200-fold reduction (from 100% ionisation) must be taken into account when calculating the MoS.

The Table above can also be used to distinguish between metallic and ionic silver in the data presented, where the SCCS requires a distinction between metallic and ionic silver. MicroSilver BG used in cosmetic formulations will have a metallic silver part of about 99.5% and ionic silver of max. 0.5%.

SCCS comment

From the provided information, the SCCS agrees that a 0.5% ionization rate for silver represents a conservative estimate of the release of ionic silver in cosmetic products. The SCCS will therefore use this value in the calculation of SED from Dermal Absorption of silver ions from the product categories mentioned in Table 2.

3.1.4. Function and uses

From SCCS/1665/24:

According to the Applicant, the microparticles offer a highly biocompatible depot of pure silver that provides sustainable release of silver ions. The sponge-like particle structure forms a relatively high surface area, compared to usual silver powder, that allows the sustainable generation of silver ions at low concentrations over a longer period of time.

3.2 TOXICOKINETICS

3.2.1 Dermal / percutaneous and oral absorption

3.2.1.1 Dermal absorption of microsilver particles

As follow-up on the publication of the SCCS Opinion SCCS/1665/24, the same Applicant submitted a new dossier with new dermal penetration studies on micron-sized particulate Silver.

A) *Ex vivo* dermal penetration study with porcine and human skin using using ToF-SIMS analysis (from Applicant's submission, abridged by the SCCS)

Guideline	No specific test guideline. Well established peer-reviewed published model (Jacobi 2007, Lademann 2010)
Test system	Excised porcine ear skin and human abdominal skin
Test substance	MicroSilver BG
Test formulation	Commercial cream containing 0.3% MicroSilver BG (Multilind® MicroSilver Cream, Stada, Germany)
Batch	Not specified
Purity	Not specified
Route	Topical application
Application area	2 × 2 cm (4 cm ²)
Application technique	Spreading formulation evenly on skin within defined area; barrier applied to prevent lateral spread
Dose of test formulation	2 mg/cm ² cream corresponding to 0.006 mg silver/cm ²
Removal of test formulation	No washing; formulation left on skin for duration of exposure
Exposure time	24 hours
Number of donors	Porcine skin: 6 animals (only 5 samples could be used) Human skin: 4 donors (female)
Skin thickness	Human skin: approx. 1.5 cm (post fat removal) Porcine skin: full thickness
Sampling	After 24 hours, treated and untreated areas excised and sectioned

Sample processing	Snap-frozen, cryo-sectioned (10 µm), mounted on ITO-coated slides
Analytical method	Time-of-flight secondary ion mass spectrometry (ToF-SIMS) (Raab 2024, Munem 2021, Hagvall 2021, Veith 2018, Nygren 2005)
Quality control	Academic research setting (Charité Berlin); ToF-SIMS was performed in an accredited laboratory (Tallarek 2025, Tascon GmbH)
Study period	2025

1
2 An *ex vivo* dermal penetration study was conducted to evaluate the distribution of silver in
3 skin following topical application of the test formulation for 24 hours. Two skin models were
4 used for the purpose: porcine ear skin and human abdominal skin. Following exposure,
5 prepared skin sections were analysed using Time-of-Flight Secondary Ion Mass Spectrometry
6 (ToF-SIMS) to visualise and semi-quantitatively assess the presence and localisation of silver
7 in the *stratum corneum* and deeper skin layers.

8 Porcine ears were obtained post-mortem from six-month-old German domestic pigs at a local
9 slaughterhouse, considered waste material from the food chain. The experimental use of the
10 tissues was authorised by the Commission of Consumer Protection and Agriculture, District
11 Dahme-Spreewald, Germany. To minimise post-mortem changes, the skin was processed
12 within 24 hours of slaughter. After rinsing under cold tap water, the ears were dried, stored
13 in a hydrated chamber at 8 °C, and prepared by shaving the dorsal surface. Two 2×2 cm areas,
14 treatment (Area S) and untreated control (Area U), were marked on the external surface, and
15 a barrier was applied around Area S to prevent lateral spreading of the formulation.

16 Human abdominal skin samples were collected from elective plastic surgery procedures
17 (female donors aged 37–60 years; Fitzpatrick skin types I–IV) with informed consent, in
18 accordance with the Declaration of Helsinki and approval by the Ethics Committee of the
19 Charité Berlin. The skin was disinfected with 70% 2-propanol prior to excision and transported
20 to the laboratory on the same day. Upon arrival, excess subcutaneous fat was removed to
21 achieve a thickness of approx. 1.5 cm. The samples were placed on a moistened surface to
22 prevent dehydration, and residual blood or fluid was gently cleaned using phosphate-buffered
23 saline (PBS). As with the porcine samples, two 2×2 cm regions (Area S and Area U) were
24 marked on the epidermal surface, with a barrier applied around Area S and left to dry for 30
25 minutes prior to cream application.

26 A commercially available cream (0.3% MicroSilver BG; Multilind® MicroSilver Cream, Stada,
27 Bad Vilbel, Germany) was applied to Area S of both skin models at a dose of 2 mg/cm² (8 mg
28 per 2×2 cm area) and spread evenly. The samples were maintained at room temperature for
29 24 hours to allow for skin penetration.

30 Following exposure, Area S and Area U were excised into four 1×1 cm sections using
31 cryospray-assisted cutting, snap-frozen, and stored at –20°C. Porcine samples were
32 sectioned within one month and human samples within 20 days. Vertical 10 µm cryosections
33 were prepared from embedded samples and mounted onto indium tin oxide (ITO)-coated
34 microscope slides. Anatomical orientation was marked to aid subsequent ToF-SIMS analysis.
35 Twelve sections were prepared per sample (three per slide across four slides). Slides were
36 air-dried overnight and analysed for silver localisation.

37 ToF-SIMS (Time-of-Flight Secondary Ion Mass Spectrometry) analysis was used to detect silver
38 and visualise its distribution within the skin. A focused ion beam scanned the surface pixel by
39 pixel to produce lateral distribution maps of silver ions, enabling spatial characterisation of
40 silver within the stratum corneum and up to 500 µm into deeper layers. RGB (Red, Green and
41 Blue) colour coding allowed for distinction between anatomical layers and the underlying slide.
42 Silver was observed in different potential forms, including metallic silver, ionic silver, silver
43 chloride, and silver–protein complexes.

Results

Porcine ear skin

Porcine ear skin was initially used to develop and optimise the sample preparation and ToF-SIMS analysis methodology. Still images extracted from video recordings of vertical cryosections from the first porcine skin sample are presented, highlighting two examined regions. The upper dark layer represents the *stratum corneum* (SC), followed by the epidermis and dermis. The areas marked in green in the Figure (Images available to the SCCS) were selected for further chemical analysis using ToF-SIMS.

Initial results from sample 1 (ear 1) indicated the need to establish a threshold based on untreated control skin. Nitrogen-containing hydrocarbons were found to be suitable markers for general skin tissue, while aliphatic hydrocarbons could be used to identify the SC, although this was not consistent across all sections (Images available to the SCCS).

Comprehensive analysis across all six porcine ear skin samples consistently showed that silver was confined to the SC, either on the surface or within its upper layers. No silver signals were detected in the viable epidermis or dermis (Images available to the SCCS).

Human excised abdominal skin

Human abdominal skin samples were processed and analysed in the same manner as porcine skin. RGB-overlay ToF-SIMS images from four human donors are shown in Figure. (note SCCS: Figure with images available to the SCCS). For donors 2 and 4, two biopsies each were analysed.

As with porcine skin, aliphatic hydrocarbons (green) consistently marked the stratum corneum, while silver signals (red) were detected only on the surface or within the upper SC layers. No silver was observed in the viable epidermis or dermis (*blue signals*).

Conclusion

The ToF-SIMS analysis of excised porcine and human skin demonstrated that any silver penetrating from the applied 0.3% MicroSilver BG cream remained restricted to the skin surface and the upper layers of the *stratum corneum*. Across all analysed samples, no silver was detected in the viable epidermis or dermis. These findings suggest that MicroSilver BG does not penetrate beyond the *stratum corneum* into living skin tissue, even following a 24-hour topical exposure. The observed heterogeneous distribution of silver is likely due to its low concentration (0.3%) within the formulation.

Overall, the finding of the study supports the conclusion that MicroSilver BG remains localised in the outermost skin barrier and does not reach viable layers of the skin, indicating no potential for systemic exposure (Meinke, 2025).

b. Study in human volunteers

Dermal absorption and mass balance study using 0.3% MicroSilver BG™ cream in human volunteers.

Guideline	No guidelines followed; Based on the principles of OECD TG 428
Test system	Human volunteers (<i>in vivo</i> tape stripping and mass balance), skin biopsies (ToF-SIMS)
Test substance	MicroSilver BG
Test formulation	Commercial cream containing 0.3% MicroSilver BG (Multilind® MicroSilver Cream, Stada, Germany)

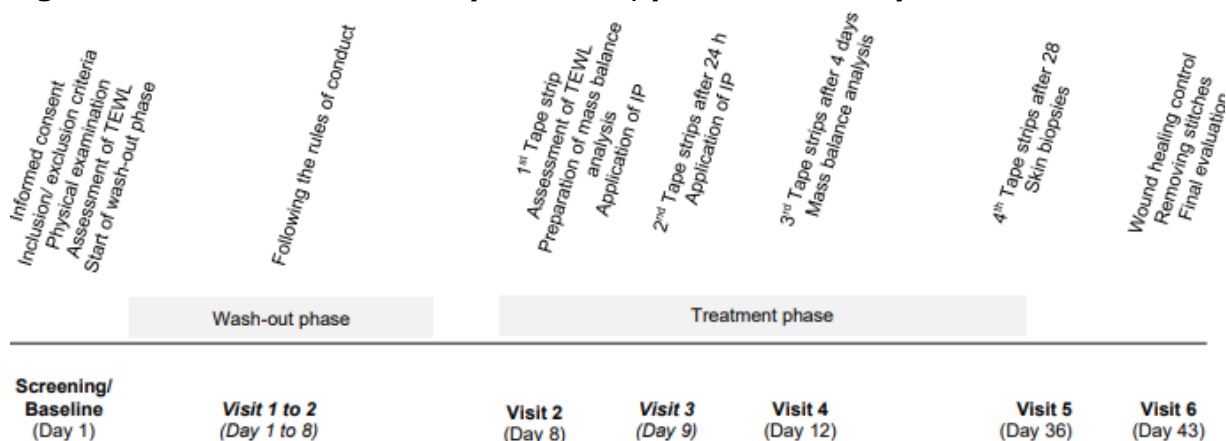
Scientific advice on Silver (CAS/EC No. 7440-22-4/231-131-3) used in cosmetic products

Batch	Not specified
Purity	Not specified
Route	Topical application
Application area	3.5 × 11.5 cm (40.25 cm ²) on lower abdomen
Application technique	Product applied once daily for 28 days at 2 mg/cm ²
Dose of test formulation	Nominal dose 2 mg/cm ² cream corresponding to 6 µg silver/cm ² (for tape stripping area); in mass balance setup, 47 mg/cm ² cream (approx. 142 µg MicroSilver BG™)
Removal of test formulation	Formulation was left on skin for duration of exposure. For sampling mild cleaning of the test area using a skin disinfectant was conducted
Mass Balance	Filter disc with Finn® chamber and occlusive Scanpor® plaster to prevent mechanical removal (Day 4, worst-case scenario for absorption)
Exposure time	24 hours, 4 days and 28 days
Number of subjects	10 healthy adults (7 females, 3 males; aged 42–59 years)
Skin thickness	Full thickness
Sample processing for Tape Stripping	Tape Stripping Study Day 9 (V3)/ 1 day after application and Study Day 36 (V5)/ 28 days after application, 60 D-Squame® discs (12 mm diameter) per site: • Pool 1: Tapes 1–30 • Pool 2: Tapes 31–40 • Pool 3: Tapes 41–50 • Pool 4: Tapes 51–55 • Pool 5: Tapes 56–60 Study Day 12 (V4)/ 4 days after application (mass balance analysis), 60 D-Squame® discs (12 mm diameter) per site: • Plaster and filter disc collected separately • Pool 1: Tape 1 • Pool 2: Tape 2 • Pool 3: Tapes 3–30 • Pool 4: Tapes 31–40 • Pool 5: Tapes 41–50 • Pool 6: Tapes 51–55 • Pool 7: Tapes 56–60
Sample processing for punch biopsies	4 mm punch from creamed area at Day 36
Analytical method	ICP-MS for -D-Squame stripping discs and mass balance components; ToF-SIMS for biopsy samples
GCP	Yes. The study was performed in compliance with ICH Good Clinical Practice.
Study period	ToF-SIMS and ICP-MS was performed in accredited labs 2025

A dermal absorption study was conducted to evaluate the skin penetration behaviour of silver following repeated topical application of a cosmetic cream containing 0.3% MicroSilver BG™ in healthy human volunteers. The study aimed to assess the extent of silver penetration beyond the stratum corneum and to determine the impact of repeated application on skin barrier integrity.

This monocentric, open-label, interventional study was carried out at the IUF–Leibniz Research Institute for Environmental Medicine (Düsseldorf, Germany) with ten healthy adult volunteers (7 females, 3 males; aged 42–59 years). All participants underwent the full 28-day treatment period, and all assessments (tape stripping, mass balance, and biopsy sampling) were performed on this same panel. After a 7-day washout period, participants applied the test formulation once daily for 28 consecutive days to a marked skin area (3.5×11.5 cm) on the lower abdomen, at a standardised dose of 2 mg/cm². To avoid site interference and allow for multiple sampling methods, separate adjacent areas were pre-designated for tape stripping, mass balance, and biopsy collection. A schematic overview of the study timeline and sample collection points is presented in Figure A.

Figure A: Overview of the study timelines, phases and analysis.



Tape stripping and ICP-MS analysis:

Tape stripping was performed at three time points to assess silver penetration in the *stratum corneum* over time: Day 9¹ (Visit 3, corresponding to 1 day after the last application on Day 8), Day 12 (Visit 4, 4 days after last application on Day 8), and Day 36 (Visit 5, 8 days after the final application on Day 28). D-Squame® discs (12 mm diameter) were applied using a standardised pressure device (D500) and removed with tweezers. A total of 60 strips per site were collected to ensure full removal of the stratum corneum. Collected discs were stored in 12-well plates and analysed for silver content via Inductively Coupled Plasma Mass Spectrometry (ICP-MS; NexION 300D, PerkinElmer).

To enable quantification above the LOQ/LOD, strips were pooled by subject and time point. Each pooled sample was weighed (± 0.1 mg), digested in 20 mL of 65% nitric acid (Suprapur®) at 150°C for 30 minutes, cooled, diluted to 100 mL with ultrapure water, and measured by ICP-MS.

Mass balance experiment (Day 12 – 4 days exposure, Visit 4):

On Day 12 (Visit 4), a mass balance experiment was conducted to better understand the fate of the applied silver. For this, a separate application site was used to apply the formulation to

¹ Study Days 9, 12, and 36 correspond to 1, 4, and 28 days of exposure, respectively. These days are aligned with Visits 3, 4, and 5 in the study schedule.

a filter disc at a higher dose (approximately 47 mg/cm², equivalent to 142 µg MicroSilver BG™). The disc was then covered with a Finn® chamber and secured using occlusive Scanpor® tape. This setup prevented mechanical removal of the product (e.g., through clothing, washing, or abrasion) and simulated a worst-case exposure scenario.

Biopsy collection and ToF-SIMS analysis (Day 36 – 28 days exposure, Visit 5):

On Day 36 (Visit 5), i.e., directly after the last application on Day 28, a 4 mm punch biopsy was collected from a designated site on each subject following local anaesthesia. Samples were analysed using ToF-SIMS, allowing spatial mapping of silver localisation in the *stratum corneum* (SC), viable epidermis, and dermis. The analysis enabled qualitative differentiation of silver signals across skin compartments.

Results

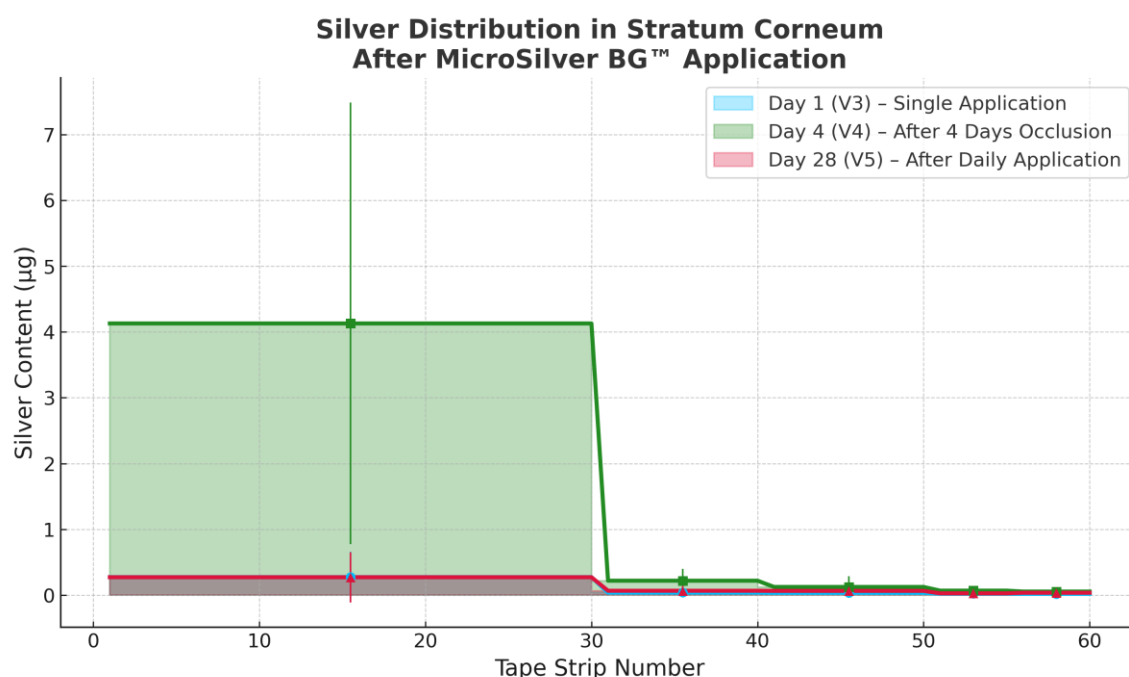
Silver content from tape strips (ICP-MS)

Quantitative analysis of silver content across sequential tape strips demonstrated a clear declining gradient from the outermost to the deeper layers of the SC. This gradient confirms limited penetration of MicroSilver BG™. The findings of the amounts of silver found in the tape after 24 hours, 4- and 28-days exposure are summarised in Table 3 below.

Table 3: Summary of silver detected (µg and % of applied silver across tape strip segments at different exposure times

	µg Silver				% of applied silver	
24 hours exposure (DAY 9, Visit 3)						
Tapes	Average	Std. Dev.	Median	IQR ²	Median	IQR
Tapes 1 - 30	0.363	0.344	0.275	0.384	4.05%	5.66%
Tapes 31 - 40	0.045	0.036	0.039	0.034	0.58%	0.50%
Tapes 41- 50	0.047	0.047	0.036	0.030	0.54%	0.44%
Tapes 51- 55	0.024	0.016	0.022	0.011	0.33%	0.17%
Tapes 55 - 60	0.029	0.030	0.021	0.039	0.31%	0.57%
4 days exposure (DAY 12, Visit 4)						
Tape 3 - 30	5.135	4.358	4.129	3.355	2.90%	2.36%
Tapes 31 - 40	0.349	0.403	0.220	0.178	0.15%	0.13%
Tapes 41- 50	0.364	0.553	0.125	0.164	0.09%	0.12%
Tapes 51- 55	0.149	0.219	0.069	0.074	0.05%	0.05%
Tapes 55 - 60	0.180	0.294	0.055	0.064	0.04%	0.05%
28 days exposure (DAY 36, Visit 5)						
Tapes 1 - 30	0.542	0.736	0.273	0.382	0.14%	0.20%
Tapes 31 - 40	0.110	0.086	0.067	0.087	0.04%	0.05%
Tapes 41- 50	0.100	0.088	0.065	0.102	0.03%	0.05%
Tapes 51- 55	0.136	0.271	0.028	0.051	0.01%	0.03%
Tapes 55 - 60	0.055	0.043	0.037	0.070	0.02%	0.04%

² Interquartile Range

Figure B: Silver distribution in stratum corneum (n = 10)**Table 4: Summary of silver detected in the deepest stratum corneum layers (tapes 56-60) following different exposure time points.**

Exposure time	Application condition	Approximate silver content in tapes 56-60	Comment
24 hours (Day 9, Visit 3)	Single application	21 ng	Baseline after first application
4 Days (Day 12, Visit 4)	Occluded application (approximately 23× dose)	55 ng	Worst-case exposure scenario (occlusion)
28 Days (Day 36, Visit 5)	Repeated daily application	37 ng	No evidence of accumulation in deeper SC layers over time

As summarised in Table 4, silver penetration in the deeper layers of the stratum corneum (tapes 56–60) varied depending on the exposure scenario. After 24 hours of daily application (corresponding to Day 9/Visit 3), the amount of silver detected in the deeper SC layers was approximately 21 ng (this measured value was in the lower part of the quantifiable range). Following a 4-day occlusive application simulating a worst-case exposure scenario (Day 12/Visit 4), silver levels increased to around 55 ng, likely due to the high dose and occlusion. However, silver remained confined to the upper SC layers. After 28 days of repeated daily application (Day 36/Visit 5), the amount of silver in the deeper SC layers was approximately 37 ng, indicating no progressive accumulation with repeated exposure.

The majority of the applied silver, approximately 280 ng on both Day 8 and Day 36, and approximately 4.1 µg on Day 12-was consistently recovered in the uppermost 30 tape strips.

The silver content in tapes beyond strip 30 was very low, indicating minimal penetration beyond the superficial layers of the SC.

Mass balance analysis (4 days exposure time – Day 12 (Visit 4)):

The mass balance evaluation showed that the applied dose per disc was 47.41 mg of cream, corresponding to approximately 142.23 µg of silver per subject. The total recovery of silver from the tape strips was 148.7 µg (approximately 104.5%) when all values were included, and 125.4 µg (approximately 88.2%) when statistical outliers were removed. These recovery rates confirm the validity of the tape stripping method and reflect good experimental consistency.

The total recovery of silver was within an acceptable range (88.2–104.5%), confirming the robustness of the sampling and analytical methodology and reliability of the study design for quantifying silver penetration and retention in the stratum corneum.

ToF-SIMS analysis:

ToF-SIMS analysis of skin biopsies (n = 10) further confirmed the localisation of silver. Silver signals were detected exclusively in the *stratum corneum*, with no signals observed in the viable epidermis or dermis. RGB overlay images enabled clear visual distinction between the SC (green), the underlying skin tissue (blue), and the silver signals (red). In biopsies 5 and 8, no silver was detected, which was likely due to microscopic inhomogeneity in the distribution of silver in the applied cream. The detected silver was present in various forms, including metallic silver, silver ions, and silver precipitates such as silver chloride (AgCl) and silver-protein complexes. All silver forms were confined to the SC, consistent with no percutaneous absorption. Silver in skin folds can be challenging in tape stripping procedures, as it could be mistakenly attributed to the lower layers of the stratum corneum. Here, ToF-SIMS analysis of biopsies has clear advantages.

Video snapshots give an overview of two biopsies from subject 1 and subject 8 as illustrated in FigureError! Reference source not found.**Error! Reference source not found.** The upper dark layer is the SC, followed by the epidermis and dermis. The green marked areas as illustrated in Figure C left are analyzed further for the different components using ToF-SIMS analysis. N-containing hydrocarbons analysis can be used to represent the skin tissue and aliphatic hydrocarbons often illustrate the SC (Figure).

Figure C: video snapshots of biopsy 1 and biopsy 8.

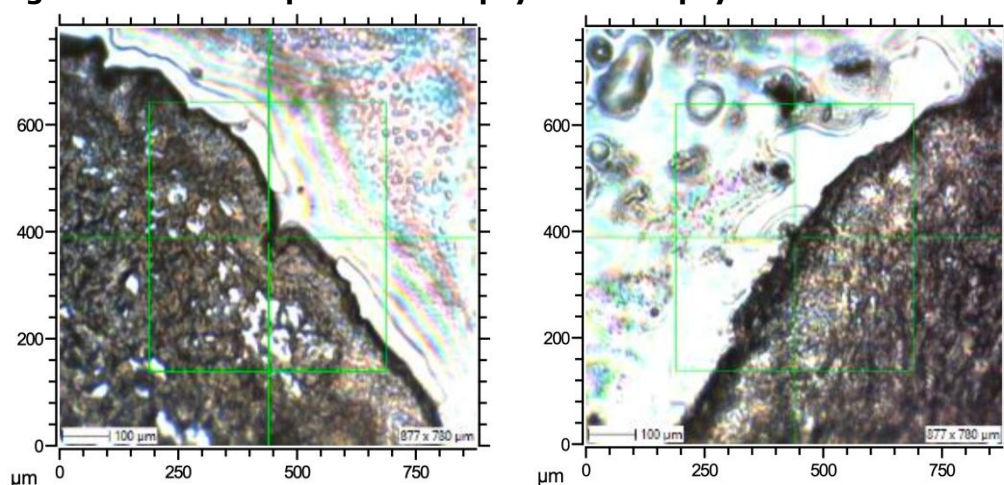


Figure D: RGB-Overlay of TOF images analyzed for Ag ⁺(red) if found, aliphatic hydrocarbons (green) and skin (blue) of two different biopsies, (image scale: 500µm x 500µm). The areas investigated are part of Error! Reference source not found..

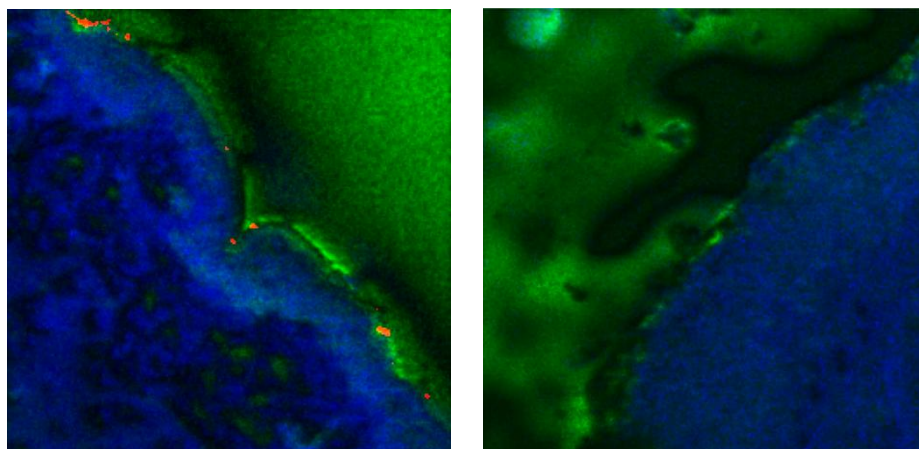
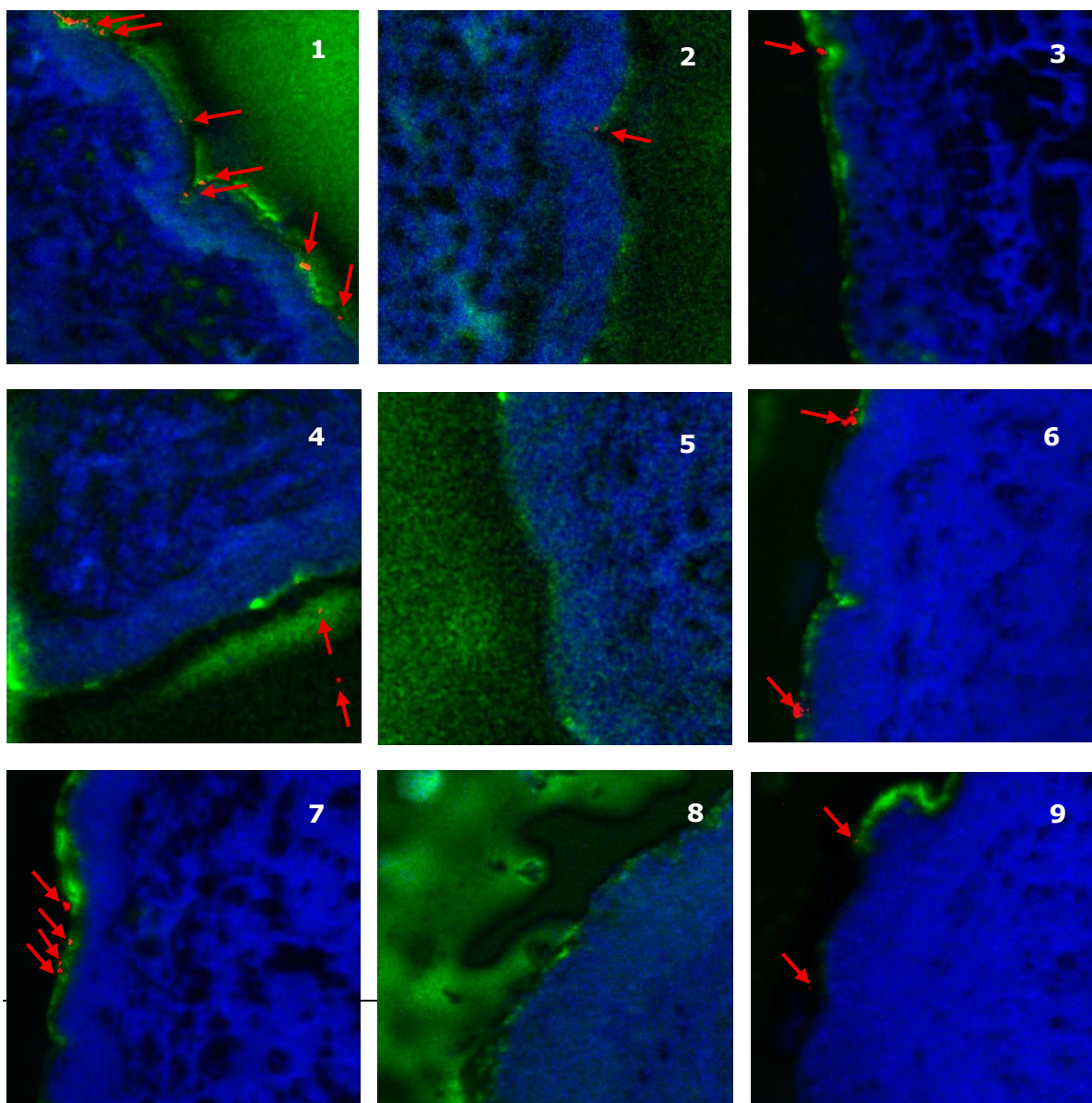


Figure E: RGB-Overlay of TOF images analyzed for Ag ⁺(red), aliphatic hydrocarbons (green) and skin (blue) of 10 biopsy samples. The number in the image indicates the number of the investigated human skin biopsy sample. The red arrows assign the silver in or on the SC, image scale: 500µm x 500µm.



Applicant's overall conclusion:

The human volunteer study demonstrated that silver from the 0.3% MicroSilver BG™ cream remained confined to the *stratum corneum* (SC) following repeated topical application. Tape stripping combined with ICP-MS analysis revealed a clear concentration gradient, with most of the silver recovered in the upper SC layers (tapes 1–30) and very low levels detected in deeper layers (tapes 31–60). No evidence of silver accumulation in the deeper SC was observed after 28 days of daily application.

ToF-SIMS analysis of human skin biopsies further confirmed these findings. Silver was detected exclusively in the SC, with no signal observed in the viable epidermis or dermis. These results were consistent across samples, although in some biopsies (e.g. subjects 5 and 8), silver was not detected at all-likely due to the low overall silver concentration in the cream (0.3%) and its heterogeneous distribution at the microscopic level.

Importantly, the Day 4 occlusion scenario, which involved approximately 23 times higher exposure under conditions known to enhance percutaneous absorption, was intended as a worst-case condition. Even under this intensified exposure, silver remained limited to superficial skin layers and did not penetrate into viable tissue. Mass balance recovery ranged between 88.2% and 104.5%, confirming the reliability and robustness of the experimental methodology.

Taken together, the data from tape stripping, mass balance, and biopsy analysis consistently demonstrate that MicroSilver BG™ does not penetrate beyond the stratum corneum and does not reach living skin layers, even after repeated application. These findings support the conclusion that MicroSilver BG™ remains localised within the outermost skin barrier and poses minimal risk of systemic exposure via dermal absorption (Krutmann, 2025)

Overall, available new studies consistently demonstrate that MicroSilver BG™ is not dermally absorbed. Silver, whether in particulate, ionic, or complexed forms, was only detected in the SC but not in the viable epidermis and dermis.

In accordance with the SCCS Notes of Guidance, substances retained within the SC at the time of sampling are not considered systemically available and therefore do not contribute to the systemic dose. Based on these findings, MicroSilver BG™ does not penetrate beyond the outermost skin layers and poses negligible risk of systemic exposure via dermal absorption.

SCCS comment

In Figure B, the SCCS noted the steep difference in the silver content in the pooled strips nr. 1 – 30 versus the strips nr. 31 and higher. While acknowledging the limitations set by the level of detection (LoD according to the Applicant: 0.03 µg/L), a pooling of strips 1-10, 11-20 and 21-30 could have been considered. Also, in view of the very small amounts observed by the overlay images in the deeper layers of the stratum corneum, the SCCS agrees that the newly submitted data indicate that Microsilver BG particles do not (or negligibly) penetrate the skin deeper than stratum corneum. Therefore, safety assessment of Microsilver BG will be focused on the systemic exposure to silver ions – not microparticulate silver.

3.2.1.2 Dermal and oral absorption of silver ions

The SED from dermal and oral exposure (see Table 5 and 6 in Chapter 3 below) will be based on the dermal and oral absorption of silver ions from the respective product categories with concentration levels listed in SCCS/1665/24.

For silver ions, a dermal absorption of 0.1% will be used, similar to the dermal absorption of silver ions already estimated in previous SCCS Opinions (SCCS/1577/16, SCCS/1650/23).

For oral exposure, the product categories and use amounts with concentration levels listed in SCCS/1665/24 will be the basis of the oral exposure. For these product categories, an oral absorption of 5% will be applied.

3.2.2 Inhalation

For inhalation exposure, the Applicant had supplied data in its original submission (17-10-2023), see Appendix.

While in that submission only deodorant spray was considered, these data seem to be related to pump-spray and not to propellant-based spray. Upon a query from the SCCS, the Applicant confirmed that Microsilver BG will not be used in propellant-based spray applications.

For the face refresh spray product, the SCCS has used the same daily amount (1540 mg) as for face cream, following the Applicant's previous submission.

Occupational inhalation limits, timeweighted for 8 hours, are set by NIOSH and by MAK at 0.01 mg/m³. (ref: MAK 2025, NIOSH 2021), whereby the NIOSH limit applies to metal dust, fume, and soluble compounds.

The SCCS performed its own calculations of the SED and for Microsilver BG particles the LED: see Tables 7 and 8 in Chapter 3.3 Exposure Assessment.

3.3 EXPOSURE ASSESSMENT

Calculation of SED/LED

In the absence of a probabilistic exposure assessment, the SCCS will use a (conservative) deterministic approach for the aggregated SED. This is based on the concentrations of Microsilver BG in the products listed in SCCS/1665/24 that had been submitted by the Applicant.

(I) Dermal

Table 5: Systemic Exposure doses (SED) of silver ions for dermal products (as listed by the Applicant), with an ionization of 0.5% and a dermal absorption of 0.1%

Product category	Product subtypes	Retention factor	Eproduct (mg/kg bw/d)	Intended use concentration (%)	External dermal dose of Ionised Ag @ 0.5%	SED (µg/kg bw/d)

Scientific advice on Silver (CAS/EC No. 7440-22-4/231-131-3) used in cosmetic products

					(µg/kg bw/d) **	
Skin care	Face cream	1	25.67	0.2	0.26	0.00026
	Face cream face tonic	1	25.67	0.1	0.13	0.00013
	Face cream ant-redness	1	25.67	0.2	0.26	0.00026
	Face cream anti pimple	1	25.67	0.2	0.26	0.00026
	Face refresh spray	1	25.67	0.1	0.13	0.00013
	Hand cream	1	36	0.2	0.36	0.00036
	Body lotion	1	130.33	0.05	0.32	0.000032
Hair care	Shampoo	0.01	1.51	0.2	0.015	0.000015
Deodorant	Deodorant spray (dermal exposure)	1	11.5	0.3	0.173	0.00017*
	Deodorant non-spray	1	25	0.3	0.375	0.00037
Foot care	Foot cream	1	20	0.2	0.2	0.0002
Make up	Eye shadow	1	0.33	0.2	0.003	0.000003
Men's cosmetics	After shave	1	20	0.1	0.1	0.0001
Aggregate dermal exposure						0.0024

*) not included in the aggregate, to avoid double counting of deodorant use.

**) [Ionised, Ag 0.5%] = [Eproduct] x [concentration] x [Retention Factor] (and include %->fraction factors) [SED] = [Ionised, Ag 0.5%] x [dermal absorption]

(II) Oral

Table 6: Systemic exposure doses (in µg/kg bw/d) of silver ions from exposure to oral products (as listed by the Applicant), applying an ionization of 0.5% and an oral absorption of 5%.

Product category	Retention factor	Eproduct (mg/kg bw/d)	Intended use concentration	External oral dose of Ionised Silver (µg/kg bw/d)	SED (µg/kg bw/d)
Lip balm	1	0.95	0.2%	0.0095	0.00047
Toothpaste adult	0.05	2.29	0.05%	0.0057	0.00028
Toothpaste children	0.4	9.22	0.05%	0.023	0.0015
Mouthwash	0.1	36.03	0.05%	0.09	0.0045
Aggregate oral exposure					0.0052 *

*Based on adult-use

(III) Inhalation of silver ions

The SCCS used a 2-box model, as described in its Notes of Guidance. In the absence of data for the respirable fraction (in this case set at 100%) and applying an ionisation rate of 0.5%, the SCCS will base the SED on the potential amount inhaled as a worst-case scenario.

Table 7: Systemic exposure doses (in µg/kg bw/d) of silver ions from inhalation exposure (see also Appendix).

Product category	Daily amount by application (mg/kg bw/d) 1)	Airborne fraction	Proportion in non-propellant	Potential amount available for inhalation, adjusted for a ionisation of 0.5% (µg/kg bw/d)	Potential amount inhaled in near-field 2)	Potential amount inhaled in far-field 3)	Retention factor	SED (µg/kg bw/d)
Deodorant spray – pump-based	93.7	0.2	0.003	0.28	0.0073	0.0037	0.75	0.0082
Face refresh spray, pump-based	25.67	0.2	0.001	0.025	0.00065	0.00034	0.75	0.00074
Aggregated								0.0089

1) Notes of Guidance 12th rev

2) Potential amount available / volume box 1 1000L x breathing rate 13 x time 2 mins

3) Potential amount available / volume box 2 10000L x breathing rate 13 x time 10 mins

(IV) Inhalation of microsilver particles

The SCCS has performed its own calculations for a local exposure dose to the lungs (Table 8), using the 2-box model described in its Notes of Guidance. In the absence of product-specific data and because the particle size of Microsilver is below the cut-off point of 5 µm, the respirable fraction will conservatively be set at 100%.

Table 8: Local External Dose of silver particles from inhalation exposure (pump-spray).

Product category	Daily amount by application per day (mg) 1)	Airborne fraction	Proportion in product	Potential amount available for inhalation (mg)	Potential amount inhaled in near-field 2)	Potential amount inhaled in far-field 3)	Retention factor	Pulmonary deposition factor 4)	Local External Dose (mg)
Deodorant spray – pump-based	6540	0.2	0.003	3.92	0.10	0.05	0.75	0.2	0.022
Face refresh spray, pump-based	1540 5)	0.2	0.001	0.31	0.008	0.004	0.75	0.2	0.002
Aggregated									0.024

1)Notes of Guidance 12th rev

2)Potential amount available / volume box 1 1000L x breathing rate 13 x time 2 mins

3)Potential amount available / volume box 2 10000L x breathing rate 13 x time 10 mins

4)Based on the Multiple-path Particle Dosimetry (MPPD) model

5)Based on daily amount face cream

In view of the conservative approach, the relatively low acute inhalation toxicity, the 8-hour time-weighted occupation exposure limits of 0.01 mg/m³ air for metal dust, fume, and soluble compounds (NIOSH 2021, also listed in MAK 2025) and the 90-day NOAEC of 0.11 – 0.75 mg/m³ air observed for the more toxic smaller nanoparticles (Hadrup 2020) , the SCCS considers this calculated LED as not being of safety concern for the cosmetic products users.

3.4 TOXICOLOGICAL EVALUATION

3.4.1. Irritation and corrosivity

From SCCS/1665/24:

ECHA-RAC considers that no classification for skin corrosion/irritation and eye irritation is warranted.

3.4.2 Skin sensitisation

From SCCS/1665/24:

Following its Opinions on a Silver containing packaging material (SCCS/1577/16) and on Silver Zinc Zeolite (SCCS/1650/23), the SCCS regards the risk of sensitisation from exposure to Silver as negligible.

3.4.3 Acute toxicity

See SCCS/1665/24.

3.4.4 Repeated dose toxicity

Taken and abridged from SCCS/1665/24:

The SCCS has reservations regarding the Applicant's choice of the PoD of 0.014 mg Silver/kg bw/day. This is further explained in 3.5 (Safety evaluation) of the SCCS/1665/24 Opinion.

The SCCS will use a NOAEL (corrected for oral bioavailability) of 0.0045 mg/kg bw/d Silver-ion equivalents for the safety evaluation. This NOAEL is the most conservative value, derived from a long-term (> 12 months) study on Silver zinc zeolite in rats (Takizawa 1992, EU CAR 2021, SCCS/1650/23).

3.4.5 Reproductive toxicity

Taken and abridged from SCCS/1665/24:

From the RAC evaluation of developmental toxicity, the SCCS will set for reproductive toxicity the most conservative NOAEL at 0.25 mg/kg bw/d silver ion equivalents, derived from a study with Silver acetate (Sprando *et al.* (2016) cited in ECHA-RAC 2022).

This is well above the NOAEL of 0.0045 mg/kg bw/d, derived from the long-term toxicity study (see 3.4.4: Repeated dose toxicity) which will be used for the overall risk-assessment in this Opinion.

3.4.6 Mutagenicity / genotoxicity

Taken and abridged from SCCS/1665/24:

The SCCS concurs with ECHA – RAC (2022) that a classification for mutagenicity is not warranted.

In its Opinion on Silver zinc zeolite (SCCS/1650/23), the SCCS stated that it agrees with ECHA/BPC/275/2021 that the genotoxic potential has been adequately investigated *in vitro* and *in vivo*. While the *in vitro* test in mammalian cells indicated a mutagenic potential of Silver zinc zeolite, there were no indications of genotoxicity in the *in vivo* studies conducted, which overrules the positive *in vitro* findings.

3.4.7 Carcinogenicity

Taken and abridged from SCCS/1665/24:

RAC considers that a classification in category 2 is not appropriate but based on the poor availability of any relevant and robust data, the information presented in the CLH dossier is considered inconclusive for the assessment of carcinogenicity. No classification for carcinogenicity is proposed due to inconclusive data. The SCCS stated in its Opinion SCCS/1650/23 that it agrees with the ECHA Opinion (ECHA/BPC/275/2021) that Silver zinc zeolite is not likely to be carcinogenic.

3.4.8 Photo-induced toxicity

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3.4.9 Human data

See SCCS/1665/24.

3.4.10 Special investigations

/

3.5 SAFETY EVALUATION (INCLUDING CALCULATION OF THE MOS)

NOAEL for silver ions, adjusted for oral absorption (see 3.4.4 Repeated Dose Toxicity):
4.5 µg/kg bw/day.

For the respective aggregate SED's, see: 3.3 Calculation of the SED.

Dermally applied products:

$$\text{MoS} = \text{NOAEL}_{\text{adj}} / \text{aggregate SED} = 4.5 / 0.0024 = 1875$$

Orally applied products:

$$\text{MoS} = \text{NOAEL}_{\text{adj}} / \text{aggregate SED} = 4.5 / 0.0052 = 865$$

Pump based spray products:

$$\text{MoS} = \text{NOAEL}_{\text{adj}} / \text{aggregate SED} = 4.5 / 0.009 = 500$$

Margin of Safety (MoS):

(conservatively) aggregated over all product categories: $4.5 / 0.0166 = 271$

For particles (see Chapter 3.3 IV): Considering the 8-hour time-weighted occupational exposure limits, and the NOAEC from a 90-day study for the more toxic smaller nanoparticles, the estimated lung exposure dose (LED) from pump spray products does not raise a safety concern for the users of these cosmetic products.

3.6 DISCUSSION

For the previous Opinion (SCCS/1665/24 the Applicant had submitted data on the dermal absorption of silver, based on an *in-vitro* study with pig skin, from which the SCCS derived a systemic exposure dose that resulted in a MoS (margin of safety) below 100 for most products. Because of this already unfavourable MoS, the SCCS did not separately consider the contribution from silver ions already present in the formulations.

In the newly submitted data, the Applicant provided new data from an *ex-vivo* study in pig and human skin and a study in human volunteers. The newly submitted data indicate that there is no or negligible dermal penetration of Microsilver BG particles, resulting in no contribution to the systemic exposure dose of Silver.

In addition, the SCCS performed its own calculations of the systemic exposure dose and subsequent risk assessment resulting from the silver ions already present in the formulations. The SCCS also estimated a lung exposure dose of Microsilver BG particles that could result from the inhalation of pump-based sprayed products, acknowledging that the calculated estimate is very conservative.

4. CONCLUSION

1. *In light of the new data provided, the previous SCCS Opinion (SCCS/1665/24) and taking under consideration the classification as toxic for reproduction Cat. 2, does the SCCS consider micron-sized particulate Silver [100 nm < particle diameter < 1 mm] safe when used up to a maximum concentration of 0.2 % in rinse-off and 0.3 % in leave-on cosmetic products?*

In the light of the new data provided, indicating that micron-sized particulate Silver does not penetrate the skin, the SCCS considers micron-sized particulate Silver [100nm<particle diameter<1 mm] safe when used up to a maximum concentration of 0.2% in rinse-off and 0.3% in leave-on cosmetic products, as specified in the Tables 5 and 6.

This Opinion does not apply to the use of micron-sized particulate Silver [100nm<particle diameter<1 mm] in propellant-based spray products because such applications were not covered by the Applicant, and therefore not assessed for safety by the SCCS.

2. *Alternatively, what is according to the SCCS, the maximum concentration considered safe for use of micron-sized particulate Silver [100 nm < particle diameter < 1 mm] in cosmetic products?*

/

3. *Does the SCCS have any further scientific concerns with regard to the use of micron-sized particulate Silver [100 nm < particle diameter < 1 mm] in cosmetic products?*

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5. MINORITY OPINION

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7. GLOSSARY OF TERMS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – from page 158

8. LIST OF ABBREVIATIONS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – from page 158

9. APPENDIX

(From the Applicant, taken from the first submission, 17 Oct 2023).

The systemic exposure dose by the inhalation route was calculated using an adapted deterministic 2- box model as described in the SCCS NoG 11th revision (SCCS, 2021). For the exposure calculations (see Table 21), use quantity of 6.1 g is considered from SCCS, 2021.

Applicant's Table 21. SED calculations after inhalation exposure

Description	Parameter	Pump spray	Unit
Amount by application ¹	A	6100 ⁹	mg/application
Fraction of MicroSilver in nonpropellant	C	0.003	(w/w)
Proportion of non-propellant in formulation	P	1	-
Airborne fraction	AF ²	0.2	-
Potential amount to be inhaled	EA (A*C*P*AF)	4	mg
First step: Near-field, 1 m ³	V1 ³	1000	L
Breathing rate	BR	13	L/min
2 min in the near field	t1	2	min
Potential amount inhaled during t1	IA1 (EA/V1*BR*t1)	0.095	mg
Second step: Far-field 10 m ³	V2 ⁴	10000	L
Breathing rate	BR	13	L/min
10 min in far-field	t2	10	min
Potential amount inhaled during t2	IA2 (EA/V2*BR*t2)	0.048	mg
Substance availability fraction	G ⁵	0.75	-
Respirable fraction	RF	0.01	-
Frequency of application	F	2	per/day
Default body weight	BW	60	kg
SED _{inhal}	(IA1+IA2) *G*RF* F/BW	0.00004	mg/kg bw/day

¹ Adjusted for the proportion of propellant to achieve a final "on-body" amount of 6100 mg

² The airborne fraction (AF) was assumed according to the SCCS NoG 11th revision (SCCS, 2021).

³ Amount of 6.1 g/day is sprayed, reported as the weight loss after use of spray (Hall *et al.*, 2007 in SCCS, 2021).

⁴ The near-field zone of the two-compartment model was assumed to have a volume V1 of 1 m³ and the duration of staying in the near-field zone t1 as 2 min.

⁵ For the far-field, a volume V2 of 10 m³ and a duration of 10 min (t2) was assumed.

⁶ The factor for substance availability G is based on Guidance from the European Commission, 1996. The respirable fraction (RF) of 0.01 is considered from (Rothe *et al.*, 2011).